I stepped into deep learning 7 years ago by building models for birdcall identification, enzyme stability estimation, vehicle motion prediction, and many more real-world challenges on Kaggle. My excitement about how deep learning connects computation and diverse aspects of our world continues to grow as I witness the potential of large language and vision models to serve as a unified engine powering machine intelligence in our complex and heterogeneous world. My research interest aims to fully realize this potential by (1) **understanding how and why models work** in increasingly open-ended settings; and (2) **building support systems** that adapt standalone models to the rugged landscape of real-world applications.

## Understanding Models and Data

Foundation language and vision models today acquire a wide range of knowledge and skills from enormous data pools. I am fascinated by (a) how models organize the information in their internal representations and (b) how data impact the models capability of acquiring the information. Ultimately, I aim to transform these insights to principled approaches that improve how we interact with models.

**Understanding from inside of models** How models can compress our entire world into a few billion parameters seems immensely beautiful to me. I am drawn to uncover how models store and use the information by understanding how they represent the information. In **SelfIE [ICML 24, First Author]**, I answered this question by repurposing the decoding capability of an LLM to decode its own hidden embedding representations into open-ended text descriptions. By laying out a map of what a model encodes at each position and each layer, SelfIE allows us to trace complex model behaviors such as reasoning on physics and susceptibility to malicious prompts. Similarly, I developed **INViTE [ICLR 24, First Author]** that repurposed decoding capabilities of CLIP to interpret its image encoder hidden layer representations into text explanations. By tracing concepts on each layer, INViTE reveals how CLIP gradually converts visual inputs in raw images to symbolic understandings in text.

After building bridges between how machines represent the world and how we see the world, I crossed the bridges to build targeted channels for humans to communicate the models. While its common to train models with human feedback on final model outputs, we extended human feedback to evaluate SelfIEs hidden embedding interpretations. This approach thoroughly erased harmful knowledge and brought stronger protections against attempts to elicit harmful behaviors. With INViTE's interpretations, users can decide whether a model should prioritize text or visual cues in images by disabling irrelevant internal connections, resolving ambiguity when images contain conflicting textual and visual information. In **EmoKnob [EMNLP 24, First Author]**, we enabled text-to-speech models with fine-grained emotion control by conducting arithmetics in the models speaker embedding space.

In the future, I hope to understand representations by understanding their relationships with model parameters. Representations flow in the models like blood, and I hope to understand how the veins the model itself transform and transport the representations. I will investigate how different model components interact to encode our diverse world.

**Understanding data** Besides the ocean of representations within models, I find the ocean of data that models float upon equally beautiful. Todays models are built on a vast ocean of data; I am drawn to how distant waves in this ocean subtly cross each other and influence model behaviors.

I started investigating data's impact with the most intuitive approach: run full training with different data mixtures. However, the enormous scale of todays data prevents abundant experiments. I was excited to find recent works of clever methods for quickly finding good data mixtures such as averaging model weights and truncating training runs. Meanwhile, I found it difficult to compare these methods since these papers conducted experiments on different setups. These observations led me to identify the core component of each work and build a unified benchmark of data mixture estimators. This process involves large-scale model training with many datasets and model configurations. I iterated through multiple systems for creating and managing experiments and learned valuable skills in building versatile yet easy-to-use model training infrastructures.

Next, I aim to understand data in emerging post-training paradigms. Post-training introduces new data forms such as synthetic labels and self-exploration that adapt models to diverse and frontier use cases. I will expand insights gained from studying pretraining data to guide informative post-training procedures and enhance models

as engines for diverse applications.

## Building Support Systems

Although studying models themselves enhances their potential as powerful intelligence engines, fully unlocking this potential and addressing diverse real-world challenges requires more than standalone models. I am interested in building support systems that adapt the bare-bone models to the complexities of our world.

**Automatic feedback** I am fascinated by how self-improving models can lead to automation in long-tail use cases where we lack abundant data or dont know a good solution. While LLMs have shown hopeful signs of self-improvement in structural domains like math and coding, we lack automatic verifiers to achieve general-purposed self-learning. I hope to build general-purposed automatic evaluations without requiring human annotations. My ongoing project aims to automatically identify disagreements across models in open-ended responses by marking response parts with diverging output probabilistic distributions. In the future, I will continue to explore general-purposed automatic feedback systems such as agents that can autonomously design external evaluation tools.

**Efficient deployment** Scaling the presence of machine intelligence in our lives requires getting answers from models quickly under economic constraints. I am interested in making out-of-factory models more efficient. This pursuit started when I learned CUDA by analyzing why `llama.cu` implements transformers faster than PyTorch and further optimized the kernel by implementing memory sharing. This experience helped me realize that there exist many under-the-hood implementation details tremendously impactful for efficient real-world applications. Following this inspiration, I immersed myself in inference efficiency challenges as I **interned at Together AI**, where I worked on enhancing speculative decoding.

I am also interested in developing learning-based methods for efficient foundation model systems. While traditional model inference handles homogeneous jobs, LLM inference handles varying input and output lengths on heterogeneous hardware. The complex memory and compute usage across scenarios make traditional scheduling heuristics less efficient. Based on **QGym [NeurIPS 24, Co-First Author]** – our study on training routing policies for queueing systems – my collaborators and I are building a differentiable simulator to adaptively learn how to route LLM inference requests across varying hardware configurations and request type distributions. I will continue to investigate how learnable systems can enable the efficient orchestration of foundation models in our heterogeneous world.

## Looking Ahead

In the long term, I aspire to become a professor dedicated to advancing scientific understanding of how machine intelligence systems operate. Additionally, I plan to pursue entrepreneurship by founding a startup that develops critical infrastructure to seamlessly integrate intelligent systems into the fabric of our society. Both goals require me to find a clear voice about what issues I deem as important and how I should shape my approach. In the past, interactions with mentors and peers helped shape my voice, and I hope to continue this journey through discussions and collaborations at PhD. The freedom to experiment with bold ideas and steer projects during PhD will also help me crystallize my intellectual perspective and research philosophy.

The works of many professors at Princeton excite me. I am particularly interested in working with **Prof. Zhuang Liu** to understand the mechanism underlying interplays between models and data. Through collaboration with **Prof. Danqi Chen**, I hope to build infrastructures for efficient model training and deployments that democratize model access. Additionally, I am excited about partnering with **Prof. Karthik Narasimhan** to design automatic feedback systems that lead to self-improving agents.


[1] **Chen, H.**, Vondrick, C., & Mao, C. (2024). *SelfIE: Self-Interpretation of Large Language Model Embeddings.* International Conference on Machine Learning (ICML).

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